

Minecarts can't pickup entities at some heights

MC-173020 Minecart

Status	Resolved / Fixed
Affected	1.15.2, 20w08a, 1.16 Release Candidate 1, 1.16, 1.16.1, 1.16.4 (+10 more)
Fixed in	22w16a
Reported	2020-02-23

STEPS TO REPRODUCE

1. Launch Minecraft 1.15.2.
2. Create a world.
3. Place a soul sand block at Y=15.
4. At Y=18, build a 3 by 3 square rail line floating around the soul sand.
5. Spawn a wither skeleton on the soul sand.
6. Place an empty minecart on the rail line.
7. Launch the minecart around the rail line.
8. Observe that the minecart does not pick up the wither skeleton.

OBSERVED BEHAVIOUR

Actual behavior

The minecart fails to detect and pick up entities whose collision boxes are in the vertical chunk section immediately below the minecart when the minecart is more than two blocks above the section boundary. The failure depends on the entities' and minecart's Y-coordinates rather than their horizontal distance.

In the reproduction above, the rail is at Y=18 and the wither skeleton is standing on the block at Y=15, across the vertical chunk-section boundary at Y=16. Although the minecart passes directly beside the skeleton and their collision boxes would otherwise overlap, the skeleton is not picked up. Moving the entire setup one block vertically changes the result, and the minecart successfully picks up the skeleton.

This behavior was observed in Java Edition 1.12.2, 1.15.2, and 1.16.1.

EXPECTED BEHAVIOUR

An empty minecart should pick up an eligible entity when it passes within the appropriate distance, regardless of the minecart's height relative to a vertical sub-chunk border. In the described setup, the minecart should consistently collect the wither skeleton, and shifting the rail line vertically by one block should not change this behavior. This should work consistently across the affected heights and versions.

ENVIRONMENT

Minecraft Java Edition Versions: 1.15.2, 20w08a, 1.16 Release Candidate 1, 1.16, 1.16.1, 1.16.4, 20w45a, 20w49a, 1.16.5, 21w05a, 21w15a, 21w16a, 1.17 Pre-release 1, 1.17.1, 21w37a, 22w13a (includes snapshots)

ORIGINAL DESCRIPTION

While testing on how to pickup ghosts with minecarts, I found that some heights are "cursed" as a minecart cannot pickup a ghost where it will at others heights. That happen on versions 1.12.2, 1.15.2 and 1.16.1.

The bug

If a minecart is more than 2 blocks up from a vertical sub-chunk border, it will never pick entities located in the vertical sub-chunk below the border.

How to reproduce

- Place a soul sand block at Y=15
- At Y=18, make a 3 by 3 blocks square rail line floating around the soul sand
- Spawn a wither-skeleton on the soul sand
- Launch a empty minecart on the rail circle

→ The minecart will not pickup the wither-skeleton, but it will if the setup is make one block offset vertically

Code analysis

From analysis on Minecraft java 1.15.1:

The bug comes from the getEntities functions in net.minecraft.world.level.chunk.LevelChunk. These functions will look at sub-chunks only 2 blocks down and up from a given collision box; so it can miss entities taller than 2 blocks.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-173020>